Xin Li

Ph.D. Student

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Research

I work on **LLM agents**, as one loop. **Evaluate**: what can models do, where do they fail, and how reliably can we tell? **Train**: how can models learn from verifiers, rewards, and other models? **Orchestrate**: how should models, tools, and verifiers work together to complete tasks reliably? Each new way of orchestrating models creates failure modes that need to be evaluated in turn. So far mostly where an answer can be checked (formal verification, mathematics, code); now, self-improvement beyond checkable answers.

Education

2025 – now **Ph.D., Nanyang Technological University**, Singapore (expected 2028)
Electrical and Electronic Engineering. Advisor: Prof. Chau Yuen (IEEE Fellow)

2018 – 2021 **M.E., Peking University**
Electronics and Communication Engineering. Advisor: Prof. Jinlong Lin. Thesis: Leveraging Geometric Regularities for Robust Visual-Inertial Odometry

2014 – 2018 **B.E., Northeastern University, China**
Measurement and Control Technology

Selected Publications

Papers where I am first, joint first (*), or second author, grouped by the research loop (evaluate, train, orchestrate), then earlier work; newest first within each group. Full list on Google Scholar.

Evaluate

NeurIPS 2026 (E&D Track) **WirelessMathBench-XL: An Auditable Benchmark for Wireless Mathematical Reasoning**
Xin Li, Mengbing Liu, Yiyang Zhu, Wenhe Zhang, Li Wei, Jiancheng An, Chau Yuen [arXiv] [Project]

ICLR 2026 **Local Success Does Not Compose: Benchmarking Large Language Models for Compositional Formal Verification**
Xu Xu*, **Xin Li***, Xingwei Qu, Jie Fu, Binhang Yuan [arXiv] [OpenReview] [Project]

TMLR 2026 **RobustMAD: Evaluating Real-World Robustness of Multimodal Small Language Models for Deployable Anomaly Detection Assistants**
Anushiya Arunan, **Xin Li**, Yan Qin, U-Xuan Tan, Nhu Khue Vuong, Xiaoli Li, Chau Yuen
[OpenReview] [Project] [Code]

ACL 2025 Findings **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**
Xin Li, Mengbing Liu, Li Wei, Jiancheng An, Mérouane Debbah, Chau Yuen [arXiv] [Project] [Code]

Train

arXiv 2026 **Many Right Answers, One Training Example: Removing Responder Noise from GRPO for LLM Recommenders**
Xin Li, Hao Jiang, Jinghao Guo, Weisi Lin, Chau Yuen [arXiv] [Project] [Code]

arXiv 2026 **Beyond Teacher Assignment: Domain-Normalized Multi-Teacher On-Policy Distillation**
Xin Li, Hao Jiang, Xin Gao, Annan Wang, Yuchen Xie, Jinghao Guo, Xingwei Qu, Yichi Zhang, Chau Yuen
[arXiv] [Project] [Code]

arXiv 2026 **ErrOPD: When Does On-Policy Distillation Transfer Teacher Errors?**
Xin Li, Hao Jiang, Annan Wang, Yichi Zhang, Chau Yuen [arXiv] [Project] [Code]

npj Wireless Tech. 2026 **Reconstructing wireless signals for low altitude networks using small language models**
Xin Li, Ran Liu, Chau Yuen [Paper]

EMNLP 2026 Industry **GraphReduce: Coverage-Preserving LLM Aggregation for E-commerce Review Insights**
Hao Jiang, **Xin Li**, Yichi Zhang, Weisi Lin [Project]

Orchestrate

arXiv 2026 **Share the Symbol, Not the Vector: Training-Free Fact Communication Across Language Models**
Xin Li, Mengbing Liu, Hao Jiang, Xin Gao, Annan Wang, Xingwei Qu, Yichi Zhang, Chau Yuen
[arXiv] [Project] [Code]

NeurIPS 2026 (E&D Track) **Measuring Collapse and Correction in Homogeneous-Panel LLM Debate**
Xin Li*, Mengbing Liu*, Chau Yuen [Project]

EMNLP 2026 Findings **Target-Local Verifier Choice in Best-of-K Reasoning Selection**
Xin Li, Hao Jiang, Weisi Lin [Project] [Code]

Earlier work

Visual-inertial SLAM: IEEE RA-L 2020 and IROS 2020 (both first author).

Experience

2026 – now **Machine Learning Engineer Intern**, *ByteDance*, TikTok Shop.
Drove the multimodal LLM review ranker for TikTok Shop across global markets: ran its full post-training pipeline (SFT → RL → on-policy distillation) and took it through online A/B tests across regions with positive results; trained the global impression-tag model on review text and video speech, selected for launch; built LLM agents that audit live review sections weekly.

2026 **Research Intern**, *Microsoft Research Asia*, Remote.
PPT agent project: rule-based checking of slides so that decks generated by AI agents follow templates and rules. Paper: AesCode: Aesthetic Code Generation with Decoupled Cross-Modal Rewards (arXiv 2026).

2024 – 2025 **Research Assistant**, *Nanyang Technological University*, Singapore.
Advisor: Prof. Chau Yuen. Built WirelessMathBench, the first benchmark for LLM mathematical modeling in wireless communications (ACL 2025 Findings), and TransPathNet for indoor radio-map prediction (ICASSP 2025).

2022 – 2024 **SLAM Algorithm Engineer (Project Lead)**, *Gausium Robotics*, Singapore.
Led five engineers on hierarchical multimodal localization (vision, LiDAR, Wi-Fi) for commercial cleaning robots: 95% localization accuracy across 100,000+ m^2, deployed on a fleet of 1,000+ robots.

2020 – 2021 **Research Intern**, *Microsoft Research Asia*, Beijing.
Mentors: Dr. Yang Liu, Dr. Yizhong Zhang. RGB-D + IMU fusion for large-scale indoor mapping; vectorized maps of 10,000+ m^2 at sub-meter accuracy.

2019 – 2020 **Research Intern**, *MEGVII Technology*, Beijing.
Mentor: Dr. Yijia He. Point-line visual-inertial odometry (IROS 2020); semi-dense mesh reconstruction at 30+ FPS.

Grants and Awards

2026 NeurIPS 2026 Top Reviewer

2025–2026 Research and compute grants: Google Gemini Academic Program (US$10,000); Modal, Cohere Labs, OpenAI (US$4,500 combined)

2025 Rohde & Schwarz Award, IEEE 6G Summit Singapore

2025 PREMIA Best Student Paper Award, finalist

2025 NTU Research Scholarship (full Ph.D. funding)

Mentoring

2026 **Chengqi Liang**, M.Sc. dissertation, NTU; went on to a Ph.D. at CUHK-Shenzhen

2026 **Yukun Jin**, B.Eng. final-year project, NTU (Wuhan University 3.5+0.5+1 programme); went on to an M.Sc. at NTU

2024 **Haoyu Xu**, M.Comp. dissertation, National University of Singapore; went on to a Ph.D. at Peking University

Invited Talks

Oct 2025 **Teaching Large Language Models Mathematical Reasoning in Wireless Communications: From Benchmarking to Efficient Training**
ISEE Youth Scholar Forum, School of Information and Electronic Engineering, Zhejiang University

Jun 2025 **WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications**
NICE Session 66: Evaluation Paper Sharing @ ICML & ACL 2025

Service

Organizer AIR4D @ IROS 2025, Workshop on Advancements for Intelligent Robotics in 4D Scenes

Reviewer NeurIPS, ICLR, ICML, AAAI, AISTATS, CVPR, ECCV, SIGGRAPH, ICRA, IROS; IEEE RA-L, ACM TOG, IEEE TNNLS